

Social learning by type-token frequency - stimulus preparation - actor focus experiment

Ansgar D. Endress
City, University of London

Abstract

Abstract (to be written)

1 Materials

1.1 Faces

We selected the stimuli from the 2,222 annotated face images in [Bainbridge et al., 2013]. For each gender and valency, we selected $4 \text{ (sets)} \times 3 \text{ (actions per set)} \times 6 \text{ (actions)} \times 3 \text{ (occurrences per set)} = 216$ faces per gender.

We selected photographs of non-famous faces with an image quality rating of at least 3.5, an estimated age between 20 and 60 years, and memorability (operationalized as [Zhang and Mueller, 2005] version of A') between the 25% and 85% percentiles. This yielded 278 and 380 photos for female and male faces, respectively. I then randomly selected the required number of faces from this initial set, with 216 faces being chosen for each gender.

The distribution of various face properties is shown in Figure 1.

1.2 Vignettes

We created 12 actions for each valence. Good and bad actions were designed to be clearly identifiable as good or bad, while avoiding actions that would be unlikely to be performed by the general population (e.g., unlike self-sacrifice or mass murder). The scenarios are given in Table 6.

Table 1: Actions

Observation	How likely are people in [city] to...	Would people in [city] consider it morally acceptable to...	Would you consider it morally acceptable to...
bad			
Noah walked to the train station and saw this person cycle past a red traffic light on their bike.	cycle past red lights?	cycle past red lights?	cycle past red lights?
Noah took the bus and saw this person lower their window to shout at the driver next to them.	shout at other drivers?	shout at other drivers?	shout at other drivers?
Noah overheard this person gossiping badly about a friend.	gossip badly?	gossip badly?	gossip badly?

Table 1: Actions (*continued*)

Observation	How likely are people in [city] to...	Would people in [city] consider it morally acceptable to...	Would you consider it morally acceptable to...
Noah saw this person park in a priority disabled parking space without displaying a blue badge.	park in priority disabled parking spaces?	park in disabled parking spaces?	park in priority disabled parking spaces?
Noah was returning a book and this person made loud noises at the library.	make loud noises at the library?	make loud noises at the library?	make loud noises at the library?
Noah walked along the canal and saw this person littering in the canals.	litter in canals?	litter in canals?	litter in canals?
Noah went to the cinema and saw this person skip the queue.	skip queues?	skip queues?	skip queues?
Noah walked along the high street and saw this person punch a police officer.	punch police officers?	punch police officers?	punch police officers?
Noah went to the bakery and saw this person smoking marijuana in front of the bakery.	smoke marijuana?	smoke marijuana?	smoke marijuana?
Noah saw this person avoid paying tube fares by jumping over the gate.	avoid paying tube fares by jumping over a gate?	avoid paying tube fares by jumping over a gate?	avoid paying tube fares by jumping over a gate?
Noah passed a farm and saw this person pillage corn from a field.	pillage corn from a field?	pillage corn from a field?	pillage corn from a field?
Noah saw this person steal a package from his friend's doorstep and drive away with it.	steal packages from a doorstep?	steal packages from a doorstep?	steal a package from a doorstep?
good			
Noah saw this person putting money in a bucket for a children's charity at the train station.	donate to children's charities?	donate to children's charities?	donate to children's charities?
Noah used the bus and saw this person offer their seat to a pregnant person.	offer their seat to pregnant people?	offer their seat to pregnant people?	offer their seat to pregnant people?
Noah passed through the main square and saw this person run to save a baby bird from being run over by a car.	save baby birds?	save baby birds?	save baby birds?
Noah visited the local cafe and saw this person buying coffee for their workplace colleagues.	buy coffee for their colleagues?	buy coffee for their colleagues?	buy coffee for their colleagues?
Noah saw this person block the cars so a child could cross the street more safely.	block streets to allow children to cross them?	block streets to allow children to cross them?	block streets to allow children to cross them?
Noah passed through a park and saw this person at the playground invite other families to share fruit.	invite other families to share fruit?	invite other families to share fruit?	invite other families to share fruit?
Noah saw this person hold the elevator door open for people rushing to get the elevator.	hold elevator doors for others?	hold elevator doors for others?	hold elevator doors for others?

Table 1: Actions (*continued*)

Observation	How likely are people in [city] to...	Would people in [city] consider it morally acceptable to...	Would you consider it morally acceptable to...
Noah dropped his wallet. This person called him to return the wallet and hand it over.	return wallets?	return wallets?	return wallets?
Noah saw this person stop reading in a cafe to help an elderly person carry heavy groceries to the bus stop.	help elderly carry heavy groceries?	help elderly carry heavy groceries?	help elderly carry heavy groceries?
Noah passed the retirement home and saw this person help an elderly person cross the road.	help the elderly cross a road?	help the elderly cross a road?	help the elderly cross a road?
Noah saw this person at the restaurant leave a generous tip to the server.	leave a generous tip to the server?	leave a generous tip to the server?	leave a generous tip to the server?
Noah got lost and when he asked this person for directions, he received a helpful answer.	give directions?	give directions?	give directions?
neutral			
Noah was passing through the park and saw this person doing kurafi.	do kurafi?	do kurafi?	do kurafi?
Noah was waiting for a taxi and saw this person walking their pet cat on the street.	walk their pet cats on the street?	walk their pet cats on the street?	walk pet cats on the street?
Noah took the bus to the city centre and saw this person running alongside cars.	run alongside cars?	run alongside cars?	run alongside cars?
Noah was at a theme park and saw this person dressed as a mascot.	dress as a mascot?	dress as a mascot?	dress as a mascot?
Noah passed through the train station and saw this person pantomiming.	pantomime?	pantomime?	pantomime?
Noah had breakfast in the office cafeteria and saw this person arriving at work by bike.	bike to work?	bike to work?	bike to work?
Noah was sitting on a bench and saw this person walk barefoot on the street.	walk barefoot on the street?	walk barefoot on the street?	walk barefoot on the street?
Noah passed in front of the post office and saw this person eating raw broccoli for lunch.	eat raw broccoli?	eat raw broccoli?	eat raw broccoli?
Noah went to the local pub and saw this person drinking a glass of wine.	drink wine?	drink wine?	drink wine?
Noah was at the footwear shop and saw this person picking their nose.	pick their nose?	pick their nose?	pick your nose?
Noah went to the harbor and saw this person walking with their swimsuit on the street.	wear swimsuits on the streets?	wear swimsuits on the streets?	wear swimsuits on the streets?
Noah saw this person at the bus stop wearing sunglasses.	wear sunglasses?	wear sunglasses?	wear sunglasses?

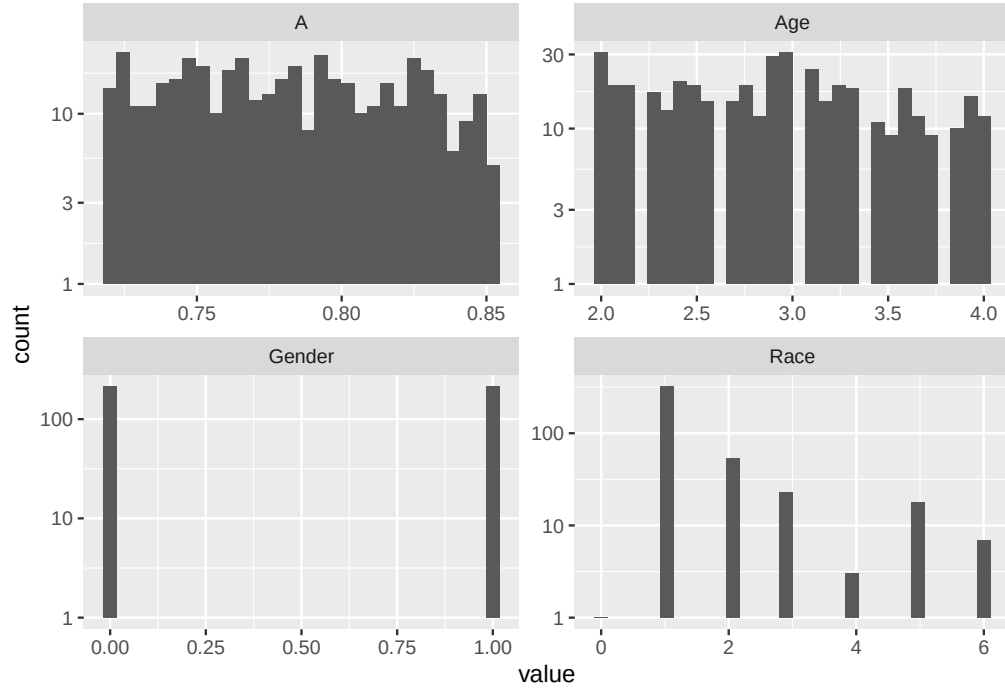


Figure 1: Distribution of various properties of the faces in the database after filtering. See main text for the filters.

1.3 City names

City names were generated using <https://www.fantasynamengenerators.com/>.

2 Design and counterbalancing

In the experiment, a traveler observed a number of actions in each city he visits. Hereafter, I refer to a “block” of trials as all actions occurring within a city, while trials are single occurrences of an action.

Each block consisted of a behavior performed 6 times by the same individual, one performed once by 6 different individuals each, and a behavior performed once by one individual. To ensure that responses were not based on prior estimates of the prevalence of individual actions, the behaviors were counterbalanced across participants.

Specifically, the 12 actions per valence were grouped into four sets of three actions. I then created three assignment conditions such that the roles of actions within blocks (high type-frequency, low type-frequency/high token-frequency, single occurrence) were counterbalanced across participants (see Table 7).

The role of the actions within the blocks was counterbalanced within participants. To maintain a reasonable experiment duration, the role of valence was counterbalanced across participants, with each participant seeing only one valence.

!!! EXPLANATION CONTINUES AFTER CHUNK !!!

Table 2: Counterbalancing of the roles of actions. Separately for each valency, actions were organized into 4 sets. Across blocks, each action of a set was presented as the high type-frequency, low type-frequency and single occurrence action.

Role	Set 1	Set 2	Set 3	Set 4
Condition 1				
High Type-Frequency	Action 1	Action 4	Action 7	Action 10
Low Type-Frequency	Action 2	Action 5	Action 8	Action 11
Single occurrence	Action 3	Action 6	Action 9	Action 12
Condition 2				
High Type-Frequency	Action 2	Action 5	Action 8	Action 11
Low Type-Frequency	Action 3	Action 6	Action 9	Action 12
Single occurrence	Action 1	Action 4	Action 7	Action 10
Condition 3				
High Type-Frequency	Action 3	Action 6	Action 9	Action 12
Low Type-Frequency	Action 1	Action 4	Action 7	Action 10
Single occurrence	Action 2	Action 5	Action 8	Action 11

Specifically, all three assignment conditions were administered to all participants. Each set of four actions thus occurred in three blocks, so that each action was seen once as high type-frequency, low type-frequency/high token-frequency and single occurrence action. The faces and place names associated with each action differed across the block. (I used the faces and placenames associated with the high type-frequency action in the original table.) This design thus yielded 12 blocks in total that appeared in random order. (In terms of testable programming, these blocks were implemented as randomblocks.)

For each participant, half of the blocks used female face and half used male faces. The correspondence between sets and genders was counterbalanced across participants. Specifically, for half of the participants, Sets 1 and 2 were presented with male faces and Sets 3 and 4 with female faces. For the remaining participants, this assignment was reversed.

This resulted in the 2 (gender/set correspondence) $\times 3$ (valence) = 6 participant groups shown in Table 8

In each block, actions were presented in a random order, with the constraint that all three types of actions (high type-frequency, low type-frequency/high token-frequency, single occurrence) appeared in the last three trials.

Table 3: Counterbalancing. The valency of the action is counterbalanced across participants. Within each valency, the gender of the face the actions are combined to is counterbalanced across participants, giving 3 groups in total.

valence	valenceSubjectGroup	genderSubjectGroup	set	faceGender
Participant Group 1				
bad	1	1	1	male
bad	1	1	2	male
bad	1	1	3	female
bad	1	1	4	female
Participant Group 2				
bad	1	2	1	female
bad	1	2	2	female
bad	1	2	3	male
bad	1	2	4	male
Participant Group 3				
good	2	1	1	male
good	2	1	2	male
good	2	1	3	female
good	2	1	4	female
Participant Group 4				
good	2	2	1	female
good	2	2	2	female
good	2	2	3	male
good	2	2	4	male
Participant Group 5				
neutral	3	1	1	male
neutral	3	1	2	male
neutral	3	1	3	female
neutral	3	1	4	female
Participant Group 6				
neutral	3	2	1	female
neutral	3	2	2	female
neutral	3	2	3	male
neutral	3	2	4	male

3 Procedure

3.1 Instructions

Participants were first told a cover story, where a student called Noah took a gap year to visit different cities. He stayed 13 days in each city and witnessed various interesting behaviors.

Participants were then instructed to estimate how widespread and morally acceptable the events in each city were, using a 4-point scale.

Participants were also encouraged to use their gut reaction.

The specific instructions are shown in Table 9.

Noah is studying journalism at university. He decides to take a gap year to explore some countries and cities that have always been interesting to him. For each of these stories, we will ask you to guess how widespread and morally acceptable the events in the story are, in the context of the country and city. We are interested in your gut reaction, so please answer with the first idea that comes to your mind after reading the story.

3.2 Attention check

Following this, participants completed an attention check where they had to select the name of the traveller from three options (Noah, Moses, Fred). Only those who passed the attention check proceeded to the experiment.

3.3 Block structure

Each trial began with a screen informing participants that Noah was about to visit a new city, and that participants were about to read about Noah’s visit in . Participants had to acknowledge this screen by pressing a button.

Then they were shown 13 trials, each involving three distinct actions (6 high type-frequency action trials, 6 low type-frequency/high token-frequency action trials, and one single occurrence action trial). Each trial featured a header indicating the day and the city name (e.g., “Day 2 in Cluakpus), a face, and a description of the action (e.g., “Noah went to the cinema and saw this person skip the queue”). While the trials were presented in random order, all three types of actions appear in the last three trials. Trials lasted 4500 ms, with a 500 ms intertrial-interval.

Finally, participants had to estimate the prevalence of each actions (i.e., “How likely are people in to ?”), its third-person acceptability (i.e., “Would *people in* consider it morally acceptable to i”), and its first-person acceptability (i.e., “Would *you* consider it morally acceptable to ?”; all emphases in original). All question were answered on a four point scale. In each trial, I first queried the high type-frequency action, then the low type-frequency action, and finally the single occurrence action.

In the pilot experiment with students, I also included the Short Dark Triad (SD3) questionnaire Jones and Paulhus [2014], which was removed in the main experiment.

A demo of a trial can be found at XXXX.

4 Stimulus generation code for testable.

For each action, we first generate we add 6 faces to each action. We thus add 12 new columns to the data frame.

We next add the set id and number within sets

We first initialize the trial file according to

randomBlock	type	stim	option
1 1	learn	list with 9 occurrences of the two high token frequency actions	randomize
1 1	learn	list with 1 occurrence of all three actions in a block	randomize
1 1	form	3 questions in fixed order for single occurrence action	header = or “In ”
1 1	form	3 questions in fixed order for low type frequency action	header = or “In ”
1 1	form	3 questions in fixed order for high type frequency action	header = or “In ”

!!! We might add a second condition where the order of the forms is inverted

4.1 SD3

We next add the Short Dark Triad (SD3) questions from Jones and Paulhus [2014]. The survey starts with the question

> Please indicate how much you agree with each of the following statements.

The head on each page is then

> How much do you agree with this statement?

Items should be kept in the same order.

5 Supplementary Only Information

5.1 Vignettes

Table 6: Actions

Observation	How likely are people in [city] to...	Would people in [city] consider it morally acceptable to...	Would you consider it morally acceptable to...
bad			
Noah walked to the train station and saw this person cycle past a red traffic light on their bike.	cycle past red lights?	cycle past red lights?	cycle past red lights?
Noah took the bus and saw this person lower their window to shout at the driver next to them.	shout at other drivers?	shout at other drivers?	shout at other drivers?
Noah overheard this person gossiping badly about a friend.	gossip badly?	gossip badly?	gossip badly?

Table 6: Actions (*continued*)

Observation	How likely are people in [city] to...	Would people in [city] consider it morally acceptable to...	Would you consider it morally acceptable to...
Noah saw this person park in a priority disabled parking space without displaying a blue badge.	park in priority disabled parking spaces?	park in disabled parking spaces?	park in priority disabled parking spaces?
Noah was returning a book and this person made loud noises at the library.	make loud noises at the library?	make loud noises at the library?	make loud noises at the library?
Noah walked along the canal and saw this person littering in the canals.	litter in canals?	litter in canals?	litter in canals?
Noah went to the cinema and saw this person skip the queue.	skip queues?	skip queues?	skip queues?
Noah walked along the high street and saw this person punch a police officer.	punch police officers?	punch police officers?	punch police officers?
Noah went to the bakery and saw this person smoking marijuana in front of the bakery.	smoke marijuana?	smoke marijuana?	smoke marijuana?
Noah saw this person avoid paying tube fares by jumping over the gate.	avoid paying tube fares by jumping over a gate?	avoid paying tube fares by jumping over a gate?	avoid paying tube fares by jumping over a gate?
Noah passed a farm and saw this person pillage corn from a field.	pillage corn from a field?	pillage corn from a field?	pillage corn from a field?
Noah saw this person steal a package from his friend's doorstep and drive away with it.	steal packages from a doorstep?	steal packages from a doorstep?	steal a package from a doorstep?
good			
Noah saw this person putting money in a bucket for a children's charity at the train station.	donate to children's charities?	donate to children's charities?	donate to children's charities?
Noah used the bus and saw this person offer their seat to a pregnant person.	offer their seat to pregnant people?	offer their seat to pregnant people?	offer their seat to pregnant people?
Noah passed through the main square and saw this person run to save a baby bird from being run over by a car.	save baby birds?	save baby birds?	save baby birds?
Noah visited the local cafe and saw this person buying coffee for their workplace colleagues.	buy coffee for their colleagues?	buy coffee for their colleagues?	buy coffee for their colleagues?
Noah saw this person block the cars so a child could cross the street more safely.	block streets to allow children to cross them?	block streets to allow children to cross them?	block streets to allow children to cross them?
Noah passed through a park and saw this person at the playground invite other families to share fruit.	invite other families to share fruit?	invite other families to share fruit?	invite other families to share fruit?
Noah saw this person hold the elevator door open for people rushing to get the elevator.	hold elevator doors for others?	hold elevator doors for others?	hold elevator doors for others?

Table 6: Actions (*continued*)

Observation	How likely are people in [city] to...	Would people in [city] consider it morally acceptable to...	Would you consider it morally acceptable to...
Noah dropped his wallet. This person called him to return the wallet and hand it over.	return wallets?	return wallets?	return wallets?
Noah saw this person stop reading in a cafe to help an elderly person carry heavy groceries to the bus stop.	help elderly carry heavy groceries?	help elderly carry heavy groceries?	help elderly carry heavy groceries?
Noah passed the retirement home and saw this person help an elderly person cross the road.	help the elderly cross a road?	help the elderly cross a road?	help the elderly cross a road?
Noah saw this person at the restaurant leave a generous tip to the server.	leave a generous tip to the server?	leave a generous tip to the server?	leave a generous tip to the server?
Noah got lost and when he asked this person for directions, he received a helpful answer.	give directions?	give directions?	give directions?
neutral			
Noah was passing through the park and saw this person doing kurafi.	do kurafi?	do kurafi?	do kurafi?
Noah was waiting for a taxi and saw this person walking their pet cat on the street.	walk their pet cats on the street?	walk their pet cats on the street?	walk pet cats on the street?
Noah took the bus to the city centre and saw this person running alongside cars.	run alongside cars?	run alongside cars?	run alongside cars?
Noah was at a theme park and saw this person dressed as a mascot.	dress as a mascot?	dress as a mascot?	dress as a mascot?
Noah passed through the train station and saw this person pantomiming.	pantomime?	pantomime?	pantomime?
Noah had breakfast in the office cafeteria and saw this person arriving at work by bike.	bike to work?	bike to work?	bike to work?
Noah was sitting on a bench and saw this person walk barefoot on the street.	walk barefoot on the street?	walk barefoot on the street?	walk barefoot on the street?
Noah passed in front of the post office and saw this person eating raw broccoli for lunch.	eat raw broccoli?	eat raw broccoli?	eat raw broccoli?
Noah went to the local pub and saw this person drinking a glass of wine.	drink wine?	drink wine?	drink wine?
Noah was at the footwear shop and saw this person picking their nose.	pick their nose?	pick their nose?	pick your nose?
Noah went to the harbor and saw this person walking with their swimsuit on the street.	wear swimsuits on the streets?	wear swimsuits on the streets?	wear swimsuits on the streets?
Noah saw this person at the bus stop wearing sunglasses.	wear sunglasses?	wear sunglasses?	wear sunglasses?

5.2 Counterbalancing

Table 7: Counterbalancing of the roles of actions. Separately for each valency, actions were organized into 4 sets. Across blocks, each action of a set was presented as the high type-frequency, low type-frequency and single occurrence action.

Role	Set 1	Set 2	Set 3	Set 4
Condition 1				
High Type-Frequency	Action 1	Action 4	Action 7	Action 10
Low Type-Frequency	Action 2	Action 5	Action 8	Action 11
Single occurrence	Action 3	Action 6	Action 9	Action 12
Condition 2				
High Type-Frequency	Action 2	Action 5	Action 8	Action 11
Low Type-Frequency	Action 3	Action 6	Action 9	Action 12
Single occurrence	Action 1	Action 4	Action 7	Action 10
Condition 3				
High Type-Frequency	Action 3	Action 6	Action 9	Action 12
Low Type-Frequency	Action 1	Action 4	Action 7	Action 10
Single occurrence	Action 2	Action 5	Action 8	Action 11

Table 8: Counterbalancing. The valency of the action is counter-balanced across participants. Within each valency, the gender of the face the actions are combined to is counterbalanced across participants, giving 3 groups in total.

valence	valenceSubjectGroup	genderSubjectGroup	set	faceGender
Participant Group 1				
bad	1	1	1	male
bad	1	1	2	male
bad	1	1	3	female
bad	1	1	4	female
Participant Group 2				
bad	1	2	1	female
bad	1	2	2	female
bad	1	2	3	male
bad	1	2	4	male
Participant Group 3				
good	2	1	1	male
good	2	1	2	male
good	2	1	3	female
good	2	1	4	female
Participant Group 4				
good	2	2	1	female
good	2	2	2	female
good	2	2	3	male
good	2	2	4	male
Participant Group 5				
neutral	3	1	1	male
neutral	3	1	2	male
neutral	3	1	3	female
neutral	3	1	4	female

Participant Group 6				
neutral	3	2	1	female
neutral	3	2	2	female
neutral	3	2	3	male
neutral	3	2	4	male

5.3 Instructions

Noah is studying journalism at university. He decides to take a gap year to explore some countries and cities that have always interested him. For each of these stories, we will ask you to guess how widespread and morally acceptable the events in the story are, in the eyes of the average person. We are interested in your gut reaction, so please answer with the first idea that comes to your mind after reading the story.

References

Wilma A. Bainbridge, Phillip Isola, and Aude Oliva. The intrinsic memorability of face photographs. *Journal of experimental psychology. General*, 142:1323–1334, 2013. ISSN 1939-2222. doi: 10.1037/a0033872. URL <https://www.wilmaabainbridge.com/facememorability2.html>.

Daniel N. Jones and Delroy L. Paulhus. Introducing the short dark triad (sd3): a brief measure of dark personality traits. *Assessment*, 21:28–41, February 2014. ISSN 1552-3489. doi: 10.1177/1073191113514105.

Jun Zhang and Shane T. Mueller. A note on roc analysis and non-parametric estimate of sensitivity. *Psychometrika*, 70(1):203–212, 2005. doi: 10.1007/s11336-003-1119-8.